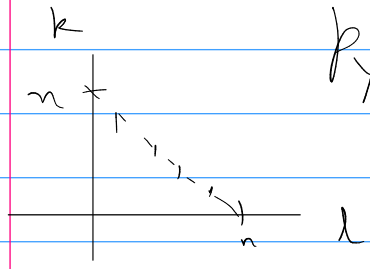
24/9/25

Ex: ① $X_1, X_2, \ldots, X_n \sim$ iid Bernoulli (p)

$Y_0 = \#\,0$'s $\quad Y_1 = \#\,1$'s

$$p_{Y_1}(k) = \binom{n}{k} p^k (1-p)^{n-k} \qquad p_{Y_0}(l) = \binom{n}{l} (1-p)^l p^{n-l}$$

$k = 0, 1, \ldots, n$ $\quad l = 0, 1, \ldots, n$



$$p_{Y_0 Y_1}(l,k) = \begin{cases} p_{Y_0}(l) = p_{Y_1}(k), & l+k=n \\ 0, & l+k \neq n \end{cases}$$

② Poisson migration

$N \sim \text{Poisson}(n)$ $\quad P(N=k) = e^{-n} \cdot \frac{n^k}{k!}$ $\quad k = 0, 1, 2, \ldots$

Given $N = m$,

$X_1, X_2, \ldots, X_m \overset{\text{iid}}{\sim}$ Bernoulli (p)

$\Downarrow$

$X_1, X_2, \ldots, X_N \sim$ iid Bernoulli (p) + $N \sim \text{Poisson}(n)$

$Y_0 = \#\,0$'s $\in \{0, 1, 2, \ldots, \infty\}$ $\sim \text{Poisson}(n(1-p))$

$Y_1 = \#\,1$'s $\in \{0, 1, 2, \ldots, \infty\}$ $\sim \text{Poisson}(np)$

$$p_{Y_1}(k) = \sum_{m=k}^{\infty} P(N=m) \cdot P(Y_1 = k \mid N=m)$$

Y_0, Y_1: indep.

$$= \sum_{m=k}^{\infty} e^{-n} \cdot \frac{n^m}{m!} \cdot \binom{m}{k} p^k (1-p)^{m-k}$$

(3) $X_1, X_2, \ldots, X_n \sim$ ⓐ iid Uniform $[0,1]$

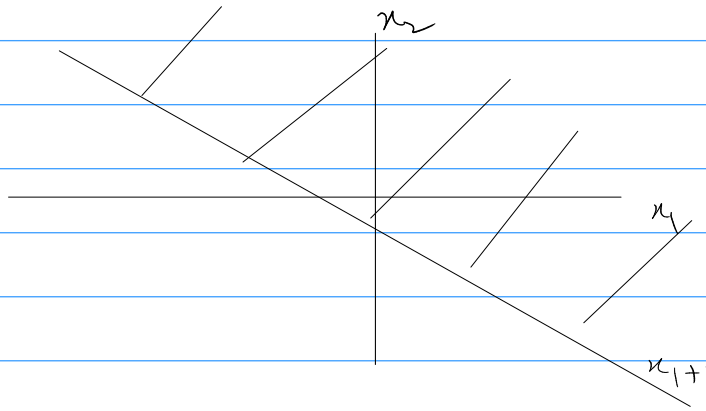
ⓑ (or) iid pdf e^{-x}, $x > 0$

$f_{X_1 X_2 \cdots X_n}(x_1, \ldots, x_n) = \prod_{i=1}^{n} f_X(x_i)$

ⓒ (or) iid pdf $\frac{1}{\sqrt{2\pi}} e^{-x^2/2}$, all x

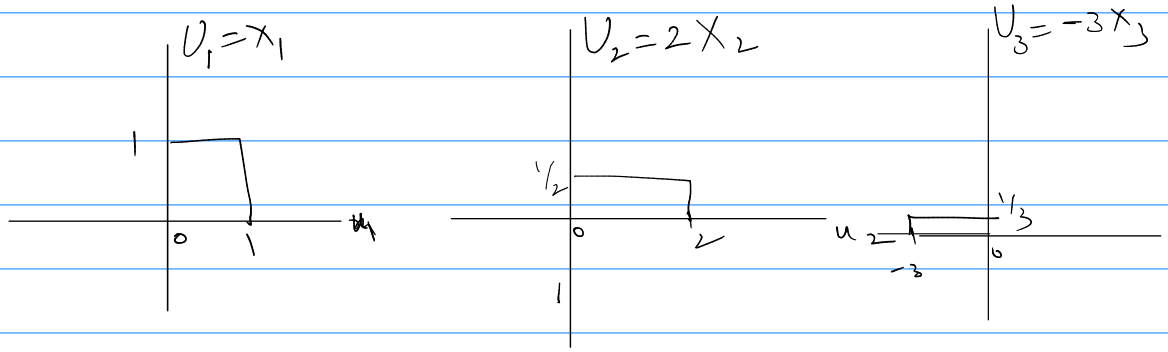
$$\Pr(X_1 + 2X_2 > 3X_3) = \iiint_{\text{Region}} f_X(x_1) f_X(x_2) f_X(x_3)\, dx_1 dx_2 dx_3$$

$$\text{Region} = \{(x_1, \ldots, x_n) : x_1 + 2x_2 > 3x_3\}$$



$$\int_{x_1=-\infty}^{\infty} \int_{x_2=\frac{b-x_1}{2}}^{\infty}$$

Uniform:



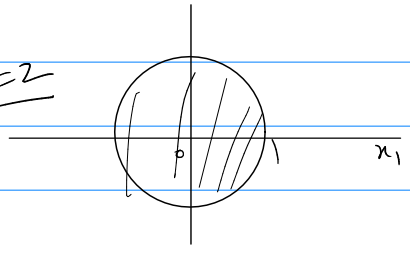
$Y = U_1 + U_2 + U_3 \sim$ pdf $f_{U_1} * f_{U_2} * f_{U_3}$

$\Pr(Y > 0)$

(4) $X_1, X_2, \ldots, X_n \sim$ Uniform { n-sphere } (unit-)

$$\{(x_1, \ldots, x_n) : x_1^2 + \cdots + x_n^2 \leq 1\}$$

$n = 2$



$$f_{X_1}(x_1) = \int_{-\sqrt{1-x_1^2}}^{\sqrt{1-x_1^2}} \frac{1}{\pi}\, dx_2 = \frac{2\sqrt{1-x_1^2}}{\pi}, \quad -1 < x_1 < 1$$

(5) $X \in \{-1, 1\}$ (1/2, 1/2)

$$Y \mid X=1 \sim \frac{1}{\sqrt{\pi}} e^{\frac{-(y-1)^2}{2}}$$

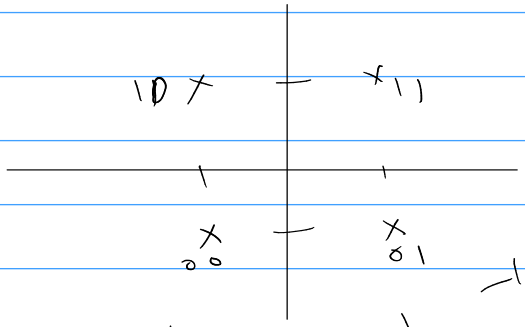
$$Y \mid X=-1 \sim \frac{1}{\sqrt{\pi}} e^{\frac{-(y+1)^2}{2}}$$

$$Pr(X=x \mid Y=y) = ?$$

$x \in \{-1, 1\}$

2D

$X_1, X_2 \sim$ iid $\{-1, 1\}$ (1/2, 1/2)



$$Y_1 \mid X_1=1 \sim \frac{1}{\sqrt{\pi}} e^{\frac{-(y_1-1)^2}{2}}$$

$$Y_1 \mid X_1=-1 \sim \frac{1}{\sqrt{\pi}} e^{\frac{-(y_1+1)^2}{2}}$$

$$Y_2 \mid X_2=1 \sim \frac{1}{\sqrt{\pi}} e^{\frac{-(y_2-1)^2}{2}}$$

$$Y_2 \mid X_2=-1 \sim \frac{1}{\sqrt{\pi}} e^{\frac{-(y_2+1)^2}{2}}$$

$$Y_1, Y_2 \mid X_1=1, X_2=-1 \sim \frac{1}{2\pi} e^{-\left(\frac{(y_1-1)^2+(y_2+1)^2}{2}\right)}$$

Y_1, Y_2 indep

$$f_{Y_1 Y_2 \mid X_1=1, X_2=-1}(y_1, y_2)$$

$$Pr(X_1=1, X_2=-1 \mid Y_1=y_1, Y_2=y_2)$$

$$= \frac{f_{Y_1 Y_2 \mid X_1=1, X_2=-1}(y_1, y_2) \cdot Pr(X_1=1, X_2=-1)}{f_{Y_1 Y_2}(y_1, y_2)}$$

$$\sum_{x_1, x_2 \in \{-1,1\}^2} f_{Y_1 Y_2 \mid X_1=x_1, X_2=x_2}(y_1, y_2) \cdot Pr(X_1=x_1, X_2=x_2)$$